

Deep Double Descent — Speaker Script

COMP34312 poster, group: Ayush Kumar, Jacob Yip, Harshita Dassani

Talk plan: 5 min each, hand-offs at 5:00 and 10:00

Time	Speaker	Owns blocks
0:00–5:00	Ayush	Header + O/P/A/C strip + 1. Hero (U-curve) + 2. Setting + 3. Definition + Hypothesis
5:00–10:00	Jacob	4. Heatmap evidence + 5. Sample-wise
10:00–15:00	Harshita	6. Mechanism + implication chain + 7. Critical reflection + 8. Practice & open questions

Iron rule: every speaker must answer Q&A on every block. The marker grades collective understanding (rubric 4c).

1. Ayush — setup and framing (0:00–5:00)

Time	Block	What to say
0:00–0:30	Header + O/P/A/C	Intro the team. Read O/P/A/C aloud verbatim. Close with: “we extend the lecture’s risk decomposition with one new measure.”
0:30–2:00	1. Hero (U-curve)	Walk Fig. 1 left to right: classical regime, critical regime (orange shading), modern regime. Dashed red curve = optimal early stopping. Hint at “larger models are not always worse.”
2:00–3:30	2. Setting	Course notation $R, \hat{R}, f_{\text{erm}}, f^*, y^*$. Excess-risk decomposition (read each underbrace). One sentence on bias–variance bundling.
3:30–4:30	3. Definition + Hypothesis	Read Definition 1 (EMC) from the boxed equation. Walk the three regime chips. Read the caveat box (“why this is a hypothesis, not a theorem”).
4:30–5:00	Hand-off	“Jacob will now show this hypothesis holds across model size, training time, and sample count.”

Ambiguities to rehearse against:

- *Interpolation threshold* (a single point) vs *critical regime* (an interval around it). Conflating them is the easiest mistake.
- *Bias–variance vs approx/est/opt*: rehearse the bridge sentence verbatim, do not improvise. Both decompose the same excess risk; bias–variance bundles approximation + part of estimation into bias.
- *What $\mathcal{T}$ contains*: $\mathcal{T} = (\mathcal{F}, \mathcal{A}, \text{epochs, regularisation, labels})$. Easy to forget the label policy.

Likely Q&A:

- *Why excess-risk decomposition not bias–variance?* Optimisation is a separate term; bias–variance can’t see epoch-wise DD.
- *What is ε and why is it heuristic?* Train-error tolerance for “fit”. Paper uses 0.1. No principled choice; the qualitative phenomenon is robust to small perturbations of ε ; the critical-interval *width* depends on it.
- *Is EMC just parameter count?* No — depends on the whole training procedure. Same architecture, longer training = bigger EMC.
- *Why hypothesis not theorem?* ε heuristic; “sufficiently smaller / larger” informal; no closed form for the critical interval; deep-net mechanism is conjectured.

2. Jacob — evidence (5:00–10:00)

Time	Block	What to say
5:00–7:30	4. Heatmaps	Same axes for both. Horizontal slice = <i>model-wise</i> . Vertical slice = <i>epoch-wise</i> . Key alignment: bright test ridge (left panel) sits <i>on</i> the train interpolation curve (right panel). Read the takeaway box.
7:30–9:30	5. Sample-wise	Fig. 3, IWSLT’14 Transformer. Light = 4k, dark = 18k. Red bars highlight the d_{model} band where $4.5\times$ more data <i>raises</i> test loss. Local non-monotonicity, not asymptotic.
9:30–10:00	Hand-off	“Harshita will explain the mechanism, our limits, and what this means for practice.”

Ambiguities to rehearse against:

- *Alignment vs causation*. “The peak is at the threshold” = alignment, not causation. The causal explanation is rigorous only for linear models (Mei & Montanari 2022 Thm 2).
- *4.5× more data hurts only in a band* ($d_{\text{model}} \approx 50\text{--}100$ on IWSLT). Not “more data is bad.”
- *Headline experiment uses 15% label noise*, but the peak appears even at 0% noise (Fig. 4a, Fig. 7). Mention proactively before being asked.

Likely Q&A:

- *Doesn't the heatmap rely on label noise?* No — same shape at 0% noise (Fig. 4a, Fig. 7). Noise just makes it more visible.
- *Why log scale on epochs?* Behaviour spans 4 orders of magnitude.
- *Could sample-wise be a finite-sample artifact?* No — the paper sweeps 4k→18k as a deliberate manipulation; the local hurt persists across multiple width values.
- *Does this contradict “more data is better”?* No — asymptotically more data wins. The local hurt is at fixed $(\mathcal{T}, \mathcal{F})$ with n moving relative to a fixed threshold.

3. Harshita — mechanism, critique, implications (10:00–15:00)

Time	Block	What to say
10:00–12:00	6. Mechanism + implication chain	Critically param: one fragile interpolant. Over-param: many interpolants + SGD’s <i>implicit bias</i> toward small-norm. Where this is rigorous: name Mei & Montanari (2022) Thm 2, Belkin et al. (2019), Bartlett et al. (2020). Walk the implication chain: smaller $n \Rightarrow \text{EMC}/n$ rises $\Rightarrow$ ensemble of small-data models stays past the per-member peak.
12:00–13:30	7. Critical reflection	Walk the four anti-claims. “Bias–variance is not wrong, it is incomplete. Bigger is not always better. Label noise amplifies but doesn’t cause. EMC is heuristic.”
13:30–14:30	8. Practice + open	Validate. Per-model escapes (weight decay, early stopping, scale). Aggregate escape (ensembling). Free escape (data augmentation). Three open questions.
14:30–15:00	Close + Q&A	Read the takeaway box. Open the floor.

Ambiguities to rehearse against:

- *SGD’s implicit bias* is proven for linear regression (min-norm interpolant). For deep nets it is intuition + partial NTK results. Always say “proven for linear, conjectured for deep.”
- *Anti-claim order*: most-to-least-damaging. Do not reshuffle under pressure.
- *Ensembling-as-DD-escape*: each ensemble member is over-parameterised relative to its own small n , so each sits past its own per-member peak; averaging cancels idiosyncratic interpolants. Paper Fig. 28.

Likely Q&A:

- *Doesn't the mechanism amount to “the optimiser is magic”?* No — for linear models it is a theorem (min-norm interpolant). For deep nets it is a conjecture. We flag this clearly.
- *Is “ensemble of small-data models” your idea or in the paper?* The component facts (DD per member, ensembling flattens the peak) are in the paper (Fig. 28). The synthesis (deliberately use small slices to keep each member over-parameterised) is our framing.
- *Should I just always ensemble?* No — ensembling is one of four escapes; the right one depends on compute and data budget.
- *What is the most important open problem?* A non-linear EMC theorem. Without it we cannot *predict* where the peak appears for a new architecture.

4. Universal handling for “I don’t know”

“That’s a good question — I don’t know the precise answer, but reasoning from definitions: [explain from EMC and Hypothesis 1]. If you have a reference in mind I’d love to read it.”

Never invent a theorem, paper, or experimental number on the fly. Rubric line 4b explicitly rewards honest uncertainty.